Practice workbook · Define a reflex and name the neurones on the receptor-to-effector path

What a reflex action is and why it matters, and naming or describing sensory, relay and motor neurones from position or role in a reflex arc.

13 question parts: 1 worked-example retry and 12 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-REFLEX-ARC-AND-NEURONE-TYPES

# 1. Question 5 · part b · 3 marks

November 2020 Paper 1

Worked-example retry · Attempt part b only.

**5** This question is about the nervous system.

(a) Describe what is meant by a reflex action.

**(3)**

(b) Diagram 1 shows a reflex arc.

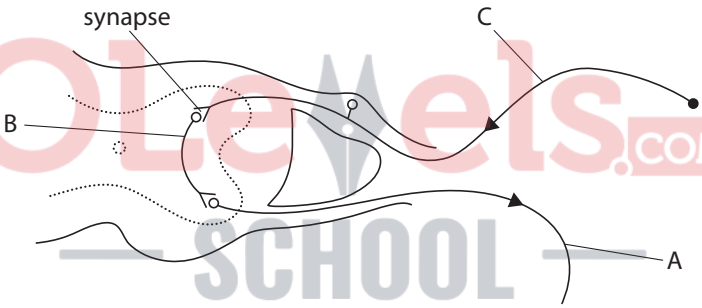



**Diagram 1**

Name the types of neurone labelled A, B and C.

**(3)**

A .....

B .....

C .....

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# 1. Question 5 · part b · 3 marks

November 2020 Paper 1

Worked-example retry · Attempt part b only.

(c) Diagram 2 shows a neurone from a healthy brain and a neurone from the brain of someone with Alzheimer's disease.

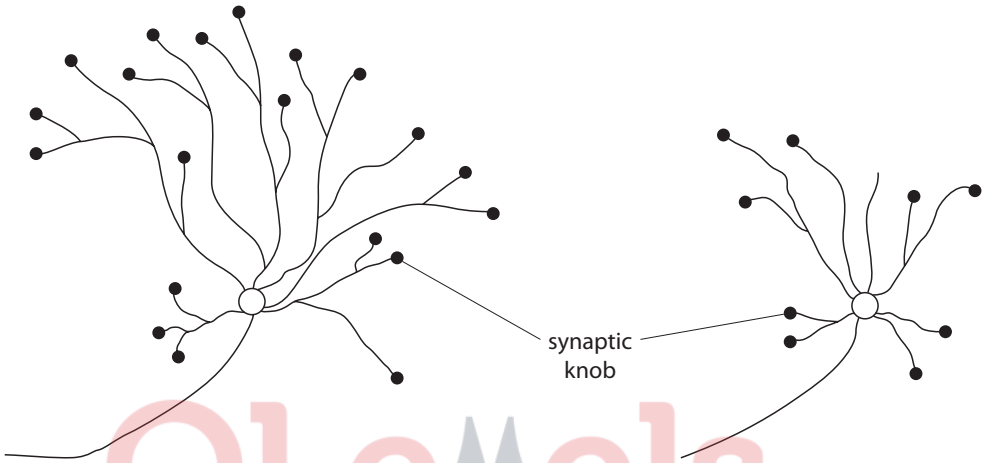



**Diagram 2**

(i) Give three differences between the two neurones. **(3)**

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2 .....

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3 .....

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## 1. Question 5 · part b · 3 marks · Working and answer

Working and answer

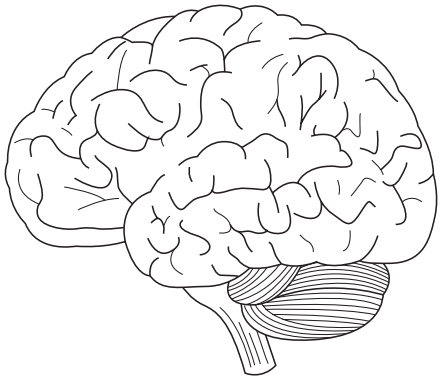
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# 2. Question 7 · part b.i · 1 mark

June 2019 Paper 1

Further practice · Attempt part b.i only.

**7** (a) The diagram shows a human brain.



Add lines labelled X, Y and Z to the diagram to show the areas of the brain that control these functions.

**(3)**

X voluntary actions

Y balance

Z breathing rate

(b) Parkinson's disease affects the cells in the brain that help to control body movement.

The affected cells are unable to communicate effectively with neurones that cause muscles to contract.

(i) Name the type of neurone that causes muscles to contract.

**(1)**

.....

(ii) Give one difference between neurones that cause muscles to contract and neurones that transmit nerve impulses from receptor organs.

**(1)**

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**20**

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## 2. Question 7 · part b.i · 1 mark · Working and answer

Working and answer

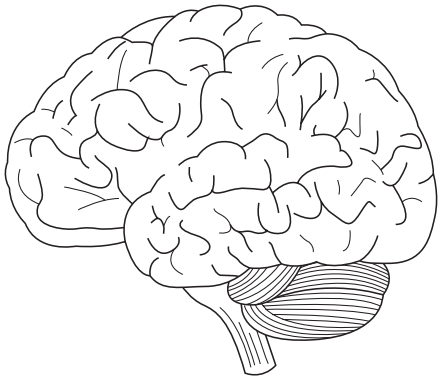
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# 3. Question 7 · part b.ii · 1 mark

June 2019 Paper 1

Further practice · Attempt part b.ii only.

**7** (a) The diagram shows a human brain.



Add lines labelled X, Y and Z to the diagram to show the areas of the brain that control these functions.

**(3)**

X voluntary actions

Y balance

Z breathing rate

(b) Parkinson's disease affects the cells in the brain that help to control body movement.

The affected cells are unable to communicate effectively with neurones that cause muscles to contract.

(i) Name the type of neurone that causes muscles to contract.

**(1)**

.....

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**(1)**

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**20**

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## 3. Question 7 · part b.ii · 1 mark

June 2019 Paper 1

Further practice · Attempt part b.ii only.

(iii) Explain how one neurone communicates with another neurone.

**(3)**

(c) Recent scientific research into the treatment of Parkinson's disease has involved the use of stem cells.

Explain how stem cells could be used to reduce the symptoms of Parkinson's disease.

**(3)**

**(Total for Question 7 = 11 marks)**

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Working and answer

# 4. Question 3 · part b · 1 mark

November 2021 Paper 2

Further practice · Attempt part b only.

**3** Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Sweating is normal when we are hot. It is part of our natural cooling mechanism. If you sweat visibly when you do not need to cool down, it can be a problem.

The sweating process is controlled by the brain, which sends signals down the nerves to the sweat glands.

Sweating is crucial to survival. We sweat to control body temperature and the sweat glands in our armpits can produce several litres of sweat in 24 hours. How much we sweat varies hugely from individual to individual and even from day to day.

Hyperhidrosis occurs when some areas of the body start sweating excessively. This affects about 1% of the population. There are two main types. The most common type is primary focal hyperhidrosis, which affects many parts of the body. There is no known cause, but it usually begins in childhood and often runs in families. Another type is secondary hyperhidrosis, which often doesn't begin until after the teenage years, and usually has an underlying medical cause. These causes can include diabetes, menopause or chronic infection. It can also be a result of eating certain foods. Stress can play a major role too.

(a) Which part of the brain controls sweating (line 4)?

**(1)**

- ☒ **A** cerebellum
- ☒ **B** cerebral hemispheres
- ☒ **C** hypothalamus
- ☒ **D** pituitary gland

(b) Which type of nerve sends signals to the sweat glands (lines 4 and 5)?

**(1)**

- ☒ **A** connector
- ☒ **B** motor
- ☒ **C** relay
- ☒ **D** sensory

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# 4. Question 3 · part b · 1 mark

November 2021 Paper 2

Further practice · Attempt part b only.

(c) Describe how sweating controls body temperature (lines 6 and 7).

**(3)**

(d) Explain the evidence in the passage that indicates primary focal hyperhidrosis is a genetic condition.

**(3)**

(e) Secondary hyperhidrosis can be caused by diabetes.

Give one risk factor for diabetes (line 16).

**(1)**

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## 4. Question 3 · part b · 1 mark · Working and answer

Working and answer

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## 5. Question 7 · part a.ii · 1 mark

June 2024 Paper 1R

Further practice · Attempt part a.ii only.

**7** (a) The nervous system sends electrical impulses to parts of the body in response to an external stimulus.

(i) Name the structure in the nervous system that detects an external stimulus.

**(1)**

.....

(ii) Name the structure that transmits electrical impulses directly to the central nervous system.

**(1)**

.....

(b) Neurones are specialised cells.

(i) Describe what is meant by a specialised cell.

**(2)**

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## 5. Question 7 · part a.ii · 1 mark · Working and answer

Working and answer

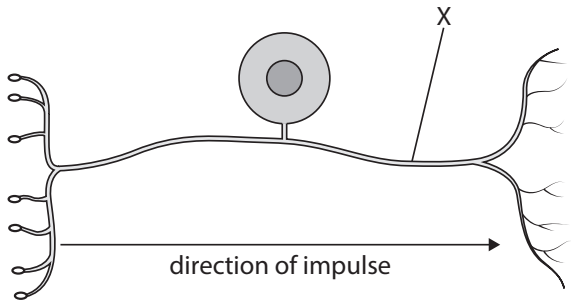
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# 6. Question 1 · part a.i · 1 mark

June 2024 Paper 1

Further practice · Attempt part a.i only.

**1** (a) The diagram shows a cell found in the human nervous system.




(Source: © Designua/ Shutterstock)

(i) The cell shown in the diagram transmits electrical impulses from a receptor cell to the central nervous system.

State the name of this type of nerve cell.

**(1)**

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(ii) Name the structure labelled X.

**(1)**

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## 6. Question 1 · part a.i · 1 mark · Working and answer

Working and answer

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# 7. Question 3 · part b.i · 3 marks

June 2025 Paper 2

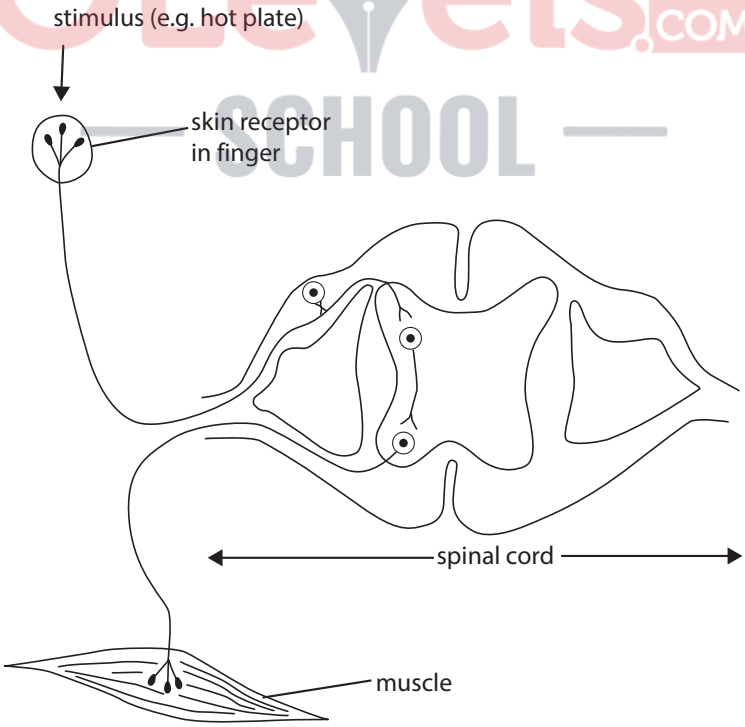
Further practice · Attempt part b.i only.

**3** (a) Describe the meaning of the term **reflex action**.

**(3)**

(b) The diagram shows a reflex arc.

The reflex arc contains three types of neurone.




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## 7. Question 3 · part b.i · 3 marks

June 2025 Paper 2

Further practice · Attempt part b.i only.

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(i) Use labels to identify each type of neurone in the diagram.

**(3)**

(ii) Draw an arrow on the diagram to show the direction of an impulse as it travels through the reflex arc.

**(1)**

(iii) State which structure in a reflex arc prevents the impulse travelling in the wrong direction.

**(1)**

**(Total for Question 3 = 8 marks)**

Working and answer

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# 8. Question 3 · part a · 4 marks

June 2022 Paper 2

Further practice · Attempt part a only.

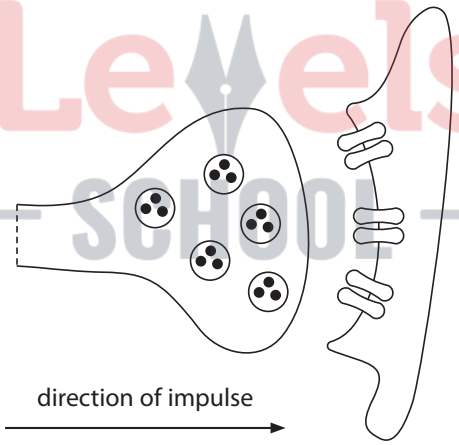
**3** (a) The nervous system is adapted to transfer information rapidly from one part of the body to another.

Complete the passage about the nervous system by giving the correct words.

**(4)**

A change in the external environment is detected by ..... in sense organs. These structures convert stimuli into electrical impulses which then travel along ..... neurones to the central nervous system. The central nervous system coordinates the incoming information using ..... neurones. These neurones then transfer the electrical impulses to ..... neurones.

(b) The diagram shows the junction between two neurones.




(i) State the name of the junction between two neurones.

**(1)**

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## 8. Question 3 · part a · 4 marks · Working and answer

Working and answer

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## 9. Question 7 · part b.iii · 2 marks

June 2024 Paper 1R

Further practice · Attempt part b.iii only.

**7** (a) The nervous system sends electrical impulses to parts of the body in response to an external stimulus.

(i) Name the structure in the nervous system that detects an external stimulus.

**(1)**

(ii) Name the structure that transmits electrical impulses directly to the central nervous system.

**(1)**

(b) Neurones are specialised cells.

(i) Describe what is meant by a specialised cell.

**(2)**

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## 9. Question 7 · part b.iii · 2 marks

June 2024 Paper 1R

Further practice · Attempt part b.iii only.

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(ii) A motor neurone is a specialised cell.

Draw a diagram of a motor neurone.

Your diagram should include these labels.

- nucleus
- cell body
- myelin sheath

**(4)**

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(iii) Describe the function of a motor neurone in the nervous system.

**(2)**

.....

.....

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## 9. Question 7 · part b.iii · 2 marks

June 2024 Paper 1R

Further practice · Attempt part b.iii only.

(c) Neurotransmitters allow different neurones to communicate with each other.

The distance between two neurones is 100 nm.

Neurotransmitters diffuse across the gap between neurones at a speed of $4.0 \times 10^6$ nm/s.

Calculate the time taken, in milliseconds, for a neurotransmitter to cross the gap between two neurones.

Give your answer in standard form.

**(4)**

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time taken = ..... ms

**(Total for Question 7 = 14 marks)**

Working and answer

## 10. Question 3 · part a.ii · 2 marks

June 2025 Paper 2R

Further practice · Attempt part a.ii only.

**3** (a) The nervous system is made up of sensory, motor and relay neurones.

(i) Draw a diagram of a motor neurone.

You should label these structures in your diagram.

- nucleus
- axon
- myelin sheath

**(4)**

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(ii) Describe the function of a motor neurone.

**(2)**

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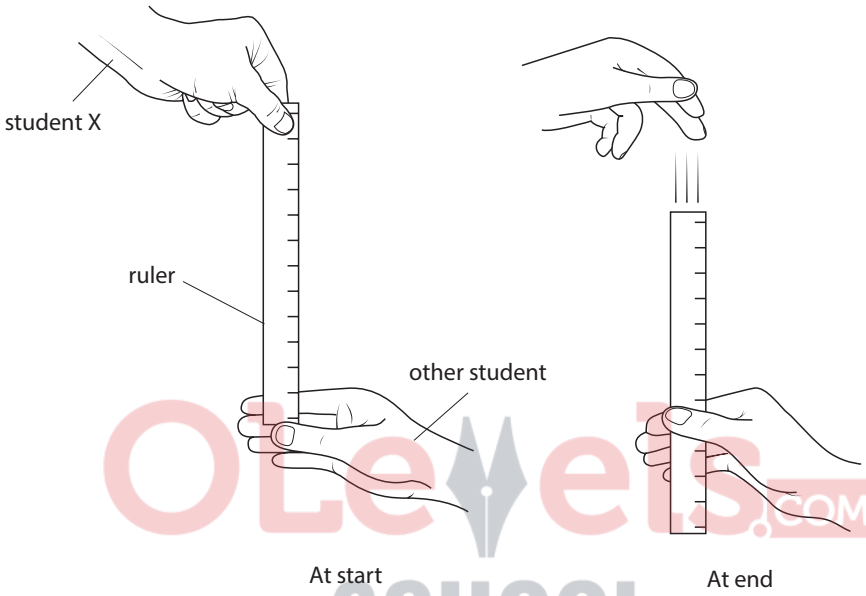
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## 10. Question 3 · part a.ii · 2 marks

June 2025 Paper 2R

Further practice · Attempt part a.ii only.

(b) Student X uses this equipment to investigate the reaction time of another student.




Student X knows that the reaction time can be calculated using this formula.

$$\text{reaction time} = \sqrt{\frac{\text{measurement from ruler}}{5}}$$

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## 10. Question 3 · part a.ii · 2 marks · Working and answer

Working and answer

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## 11. Question 5 · part a · 3 marks

November 2020 Paper 1

Further practice · Attempt part a only.

**5** This question is about the nervous system.

(a) Describe what is meant by a reflex action.

**(3)**

(b) Diagram 1 shows a reflex arc.

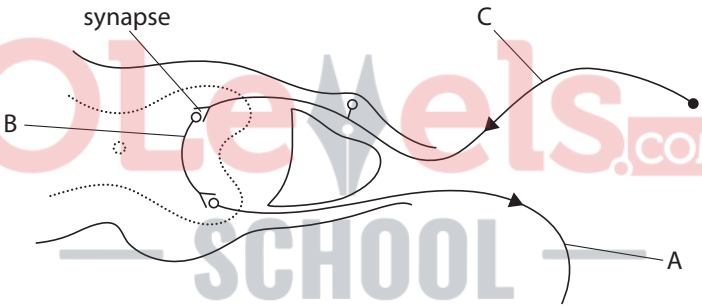



**Diagram 1**

Name the types of neurone labelled A, B and C.

**(3)**

A .....

B .....

C .....

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## 11. Question 5 · part a · 3 marks · Working and answer

Working and answer

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## 12. Question 8 · part a · 3 marks

June 2021 Paper 1

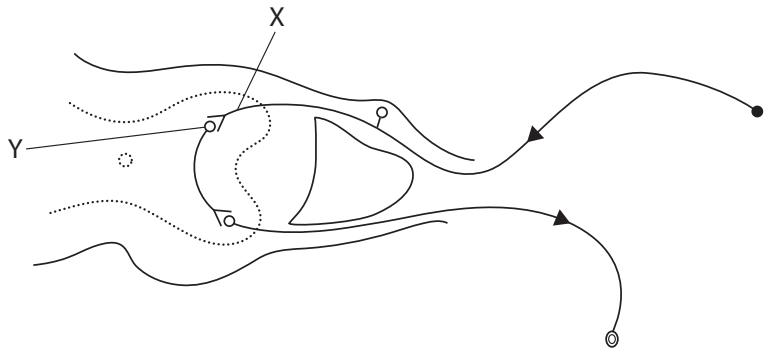
Further practice · Attempt part a only.

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**8** The diagram shows a reflex arc involved in a reflex action.




(a) Describe the function of a reflex action.

**(3)**

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(b) Using a line labelled Z, show on the diagram which part of the reflex arc would be connected to a receptor.

**(1)**

(c) Describe how the nerve impulse passes from position X to position Y in the reflex arc.

**(6)**

**(Total for Question 8 = 10 marks)**

## 12. Question 8 · part a · 3 marks · Working and answer

Working and answer

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## 13. Question 3 · part a · 3 marks

June 2025 Paper 2

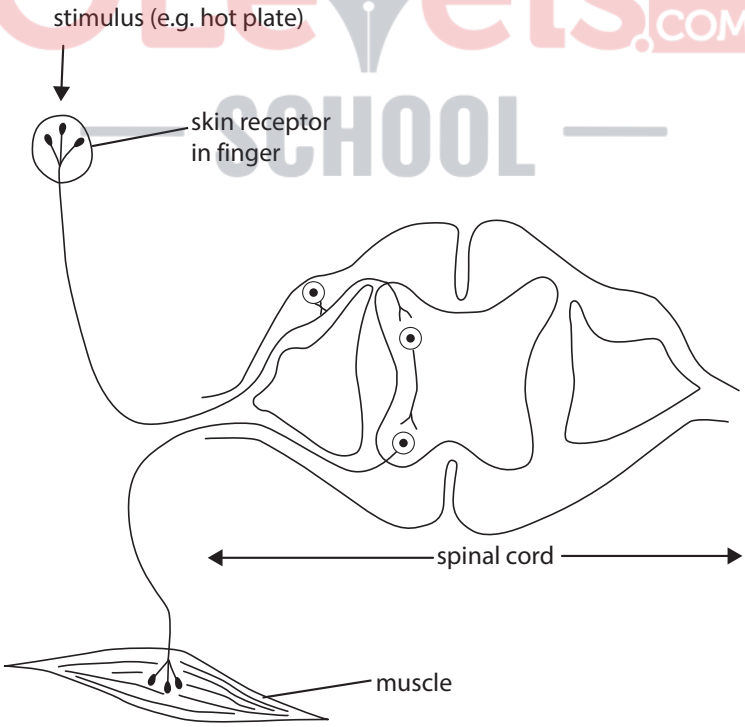
Further practice · Attempt part a only.

**3** (a) Describe the meaning of the term **reflex action**.

**(3)**

(b) The diagram shows a reflex arc.

The reflex arc contains three types of neurone.




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## 13. Question 3 · part a · 3 marks · Working and answer

Working and answer

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